DG INTERNS HUB — TECHNICAL NOTES

Topic 01: What Is SIEM (Security Information and Event Management)?

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1. Introduction to SIEM

In a modern enterprise network, hundreds of devices—including workstations, servers, firewalls, and domain controllers—generate millions of log entries every single minute. If an organization relied on security analysts to manually log into individual machines and review text files to look for malicious activity, threats would go completely unnoticed. Attackers would easily slip past defenses, resulting in prolonged dwell times and devastating data breaches.

This is where Security Information and Event Management (SIEM) technology becomes vital. As an aspiring Security Operations Center (SOC) analyst, understanding SIEM is foundational because it serves as the central nervous system of modern security monitoring. It aggregates disparate log streams into a single pane of glass, giving analysts the visibility needed to detect, investigate, and mitigate threats.

2. What is SIEM?

SIEM (Security Information and Event Management) is a comprehensive software solution that combines two primary security disciplines into a single unified platform:

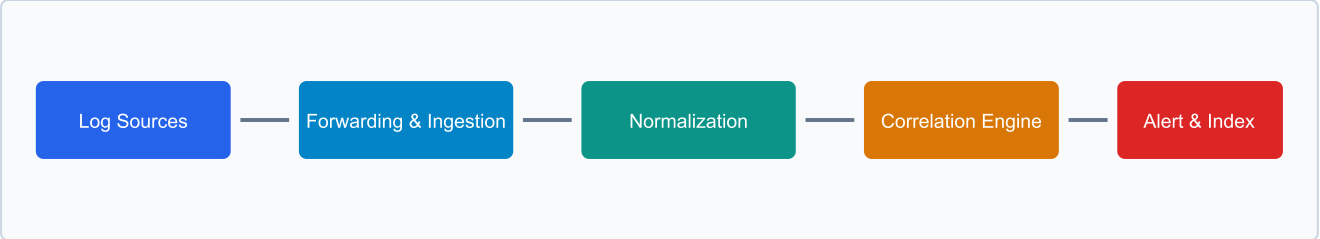
- **Security Information Management (SIM):** Focuses on collecting, analyzing, and storing log data from various systems for compliance reporting and long-term historical analysis.
- **Security Event Management (SEM):** Focuses on real-time monitoring, event correlation, analysis of security alerts, and immediate incident notification.

When combined, a SIEM collects raw log data across the enterprise, normalizes it into a common structure, correlates events in real-time, and triggers alerts when suspicious patterns emerge. Popular solutions include Splunk Enterprise, IBM QRadar, Microsoft Sentinel, and Elastic SIEM.

3. Why SIEM is Important in Cybersecurity

4. How SIEM Works

The operational lifecycle of telemetry moving through a SIEM follows a structured data pipeline:



1. **Generation:** Operating systems and applications produce events.
2. **Collection:** Agents gather logs and send them to central servers.
3. **Normalization:** Unstructured data is converted into standard key-value fields.
4. **Correlation & Analysis:** Rules compare events across sources to flag threats.
5. **Action:** Incidents are indexed for rapid search and alerts are dispatched.

5. Main Components of a SIEM System

- **Log Forwarders / Agents:** Lightweight modules installed on endpoints that collect and forward telemetry to the central engine.
- **Collectors / Receivers:** Ingestion points that securely receive log streams over Syslog, HTTPS, or API connections.
- **Parsing Engine:** Software layer breaking raw strings (JSON, XML, unstructured) into queryable schema fields.
- **Correlation Engine:** Analyzes events against rule sets (e.g., "5 failed logins within 1 minute followed by 1 success").
- **Storage & Indexer:** High-performance database optimized for fast writes and long-term historical search.
- **Dashboard Interface:** The GUI used by SOC analysts for queries, dashboards, and triage.

6. Types of Logs Collected by SIEM

- **Security Event Logs:** Authentication, privilege usage, and account changes (e.g., Windows Event IDs 4624, 4625).
- **Network Traffic Logs:** Firewall rule hits, NetFlow, DNS query logs, and Proxy connections.
- **Endpoint Telemetry:** EDR logs containing process trees, command-line parameters, and file writes.
- **Application Logs:** Web server access logs (IIS, Apache), database queries, and cloud audit trails.

7. Log Collection, Normalization, Correlation, and Analysis

The core strength of a SIEM lies in how it processes telemetry:

- **Collection:** Pulls logs from thousands of endpoints via push or pull mechanisms.
- **Normalization:** Converts disparate vendor formats into a unified schema (e.g., mapping Windows `TargetUserName` and Linux `user` both to `src_user`). This allows cross-platform querying.
- **Correlation:** Connects separate events. A single network connection to an external IP might seem benign, but correlated with a concurrent PowerShell script downloading an execution payload, it highlights an active compromise.
- **Analysis:** Searching historical indices to baseline normal behavior and isolate outliers.

8. SIEM Alerts and Incident Detection

SIEM platforms fire alerts through three primary mechanisms:

- **Threshold-Based:** Triggers when an event count exceeds limits (e.g., >10 failed logins in 30 seconds).
- **Signature-Based:** Matches known malicious indicators (IoCs) like file hashes or suspicious command flags.
- **Anomaly-Based:** Flags deviations from standard user or entity behavior baselines (UEBA).

9. Practical SOC Investigation Example

> **Scenario: Investigating a Brute-Force Attack in Splunk**
> An alert fires in the queue: *Multiple Failed Logins Followed by Successful Authentication.*

As an analyst, I begin investigating by querying Windows Security Event logs in Splunk using Search Processing Language (SPL):

```
index=windows EventCode=4624 OR EventCode=4625 user="jdoe"
| stats count by EventCode, src_ip, dest_computer_name, _time
| sort _time
```

Findings: The query reveals 12 consecutive `EventCode=4625` (Failed Logon) entries within 20 seconds originating from IP `192.168.1.150`, followed instantly by `EventCode=4624` (Successful Logon). This confirms a successful brute-force attack. I immediately escalate the ticket to isolate the workstation and force a user password reset.

10. How a SOC Analyst Uses SIEM

On a daily shift, a Tier 1 SOC Analyst interacts with the SIEM to perform essential security tasks:

- **Alert Triage:** Reviewing incoming real-time alerts, separating True Positives from False Positives.
- **Threat Hunting:** Writing custom search queries to discover quiet or undetected adversary presence.
- **Rule Fine-Tuning:** Assisting engineers in reducing alert fatigue by tweaking false-positive thresholds.
- **Incident Documentation:** Compiling event timelines, target host lists, and forensic indicators into case notes.

11. Benefits and Limitations of SIEM

Capability / Aspect	Key Benefits	Operational Limitations
Visibility & Centralization	Single pane of glass across all enterprise endpoints and network controls.	Requires significant network bandwidth and storage infrastructure.
Automation & Correlation	Detects complex multi-stage attacks in real-time across data sources.	Prone to high false-positive rates if rules are un-tuned.
Forensics & Compliance	Provides reliable historical audit trails and simplified compliance reporting.	Licensing costs scale rapidly with high daily ingestion volumes (GB/day).

12. Best Practices

- **Prioritize Log Sources:** Focus initial ingestion on high-risk assets like Domain Controllers, Firewalls, and Cloud Gateways.
- **Continuous Tuning:** Regularly evaluate fired alerts to eliminate recurring false positives.
- **Implement RBAC:** Enforce Strict Role-Based Access Control so logs are protected against unauthorized modification.

13. Conclusion

Through my study of SIEM technology during Week 4, I have learned that a SIEM is far more than a centralized repository for log storage—it is the vital intelligence engine of a Security Operations Center. By aggregating, normalizing, and correlating event streams, SIEM solutions allow analysts to detect complex attacks that would otherwise remain hidden. Understanding log pipeline mechanics and mastering query tools like Splunk are essential steps in my training as a SOC analyst.